INTERNSHIP COURSE
WEEK 1 TASK

STRATEGIC MARKET ANALYSIS & RESEARCH

Apparel & Textiles Industry with a Focus on Machine Learning

Submitted By

PANDHULA SINDUJA

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EXECUTIVE SUMMARY

The apparel and textiles industry is changing quickly because customer preferences, fashion trends, online shopping, prices and supply-chain conditions can change within a short period. For companies, one of the most difficult decisions is knowing what customers are likely to buy and how much stock should be kept available. If demand is overestimated, products may remain unsold and require markdowns. If demand is underestimated, stock-outs can lead to lost sales and dissatisfied customers.

This Week 1 study looks at how machine learning and data analytics can help apparel businesses make these decisions more effectively. The main focus is demand forecasting and inventory decision support, while other applications such as customer segmentation, product recommendations, production planning, quality analysis and sustainability are also considered.

The research is based on publicly available industry information and datasets. The proposed approach moves from industry research to data collection, data cleaning, exploratory analysis, feature engineering, customer and product analysis, forecasting, evaluation and business recommendations. The purpose is not only to build a model, but to understand how analytical results can be converted into useful decisions.

1. Introduction

Apparel and textiles is one of the most important consumer and manufacturing sectors. It includes activities from fibre and yarn production through fabric processing and garment manufacturing to wholesale, retail and recycling. The sector is highly sensitive to customer behaviour because products can have short selling periods and demand may differ by season, size, colour, price, location and current fashion trends.

The growth of e-commerce has also changed the amount of information available to retailers. Online transactions can provide detailed information about products, customers, prices, purchase dates and buying patterns. When this information is analyzed properly, machine learning can help identify patterns that may not be obvious from manual analysis.

For this reason, this research investigates the strategic role of machine learning in the apparel and textiles industry, with particular attention to demand forecasting, customer behaviour and inventory management.

2. Market Overview

2.1 Industry Overview

The apparel and textiles value chain covers fibre production, spinning, yarn manufacturing, fabric formation, dyeing and processing, garment manufacturing, wholesale, retail and post-consumer reuse or recycling. India has a broad textile ecosystem that includes cotton and other fibres, yarn, fabric, apparel, home textiles and technical textiles.

The industry research used for this project reports that the Indian textile and apparel sector is expected to move toward a US$350 billion market by 2030. It also reports FY2025–26 textile and apparel exports including handicrafts of US$35.52 billion. These figures indicate the scale of the sector and the importance of improving productivity, technology adoption, supply-chain resilience and demand understanding.

2.2 Major Current Trends

Industry Trend	What It Means	Machine Learning Opportunity
Demand volatility	Customer demand changes with seasons, prices, promotions and trends.	Demand forecasting and demand-risk analysis.
E-commerce and omnichannel retail	Digital channels create detailed transaction and customer information.	Customer analytics, recommendations and demand sensing.
Artificial intelligence and automation	AI is increasingly being considered for forecasting and operational decisions.	Predictive models, optimization and decision-support dashboards.
Sustainability	Companies face pressure to reduce waste and resource consumption.	Waste, material and lifecycle analytics.
Supply-chain diversification	Disruptions encourage businesses to build more resilient sourcing networks.	Supplier risk scoring and scenario analysis.

3. Problem Statement

Apparel businesses often have difficulty deciding how much of each product should be produced, purchased or kept in stock. Demand can change across time, product categories, customer groups and price levels. An inaccurate estimate may result in excess inventory, markdowns and waste, while an underestimate can result in stock-outs and lost sales.

The problem selected for this internship research is: How can historical sales and product/customer information be used to forecast apparel-related demand and identify products that are at higher risk of overstock or understock?

The project treats inventory optimization as a decision-support problem. Public sales datasets may not contain complete opening-stock, receipts and closing-stock information. Therefore, the research will first focus on forecasting sales demand and developing an inventory-risk framework. If genuine inventory records become available later, the framework can be extended.

4. Why This Problem Matters

- Inventory affects working capital, storage costs, markdown exposure and profitability.
- Fashion products can lose commercial value when seasons and trends change.
- Stock-outs can immediately reduce sales and may encourage customers to choose competitors.
- Better forecasts can improve procurement and production planning.
- Reducing unnecessary production and unsold stock can also support sustainability.
- The problem has a clear connection between data, business insight, operational decisions and measurable KPIs.

5. Machine Learning Opportunities

5.1 Demand Forecasting

Demand forecasting is one of the most useful applications for apparel retail. Historical sales can be combined with time, product, price and seasonal features to estimate future demand. Forecasts can then support purchasing, production and replenishment decisions.

5.2 Customer Segmentation

Customer segmentation helps a company understand that not all customers behave in the same way. RFM analysis (Recency, Frequency and Monetary value) can be used to describe customer value and activity. Clustering methods such as K-means can then be considered when the data is suitable.

5.3 Inventory-Risk Prediction

A product can be given a risk score using its demand level, demand volatility and forecast uncertainty. This can help managers identify products that need closer monitoring, more flexible ordering or different markdown strategies.

5.4 Product and Assortment Planning

Analytics can help identify which categories, sizes, colours and products have stronger demand. This can support more informed assortment planning rather than relying entirely on intuition.

5.5 Production and Supply-Chain Analytics

Predicted demand can be used as an input for production scheduling. Similar analytical methods can also be used to compare suppliers, identify supply risks and evaluate possible sourcing scenarios.

5.6 Quality and Sustainability Analytics

If inspection, return, material, energy, water or waste data are available, analytics can help identify quality problems and prioritize sustainability improvements.

6. Review of Public Research and Industry Evidence

The industry sources reviewed for this project describe a market environment affected by economic uncertainty, changing customer behaviour, trade changes, climate pressure and inventory challenges. They also point to the growing use of AI for customer discovery, forecasting, inventory prediction and production planning.

The research also shows that AI should not be viewed only as a technology project. Its value comes from connecting data and predictions to practical business decisions. For example, a forecast becomes useful when a retailer can use it to decide whether to replenish a product, reduce an order, adjust a promotion or change a planning cycle.

For the Indian context, publicly available industry information indicates strong growth potential and continued export activity. This creates an opportunity for data-driven methods to improve competitiveness, particularly in areas such as demand understanding, inventory efficiency and supply-chain resilience.

7. Research Methodology

1. Industry research: Review current apparel and textile trends, challenges and opportunities.

2. Problem definition: Define the business problem, research questions and key performance indicators.

3. Dataset discovery: Identify public datasets and record their source, scope and limitations.

4. Data preparation: Clean and standardize the selected data.

5. Exploratory analysis: Study sales, products, customers, prices, trends and seasonality.

6. Feature engineering: Create useful time, demand, product, customer and price features.

7. Segmentation: Analyze customer and product behaviour.

8. Forecasting: Develop a baseline and compare forecasting approaches.

9. Risk scoring: Convert demand and volatility information into product-level priorities.

10. Evaluation: Compare models using appropriate metrics and time-based validation.

11. Business interpretation: Translate findings into practical recommendations.

8. Potential Data Sources

Data Source	Main Information	Possible Use
UCI Online Retail	541,909 transactions with product, quantity, date, price, customer and country fields.	Demand trends, customer segmentation and product analysis.
UCI Online Retail II	More than one million transactions across two years.	Longer-term demand and time-series analysis.
Kaggle Fashion Retail Sales	Fashion-oriented transaction information, including purchase and rating fields.	Fashion preference and sales analysis.
India Open Government Data	Textile production and export information.	Indian market and category benchmarking.
UN Comtrade	International trade data by product and partner country.	Export and market comparison.

9. Data Analysis Plan

9.1 Data Cleaning

- Check duplicate transactions and identifiers.
- Convert transaction dates into a standard format.
- Check missing customer, product and price information.
- Document rules used for cancelled transactions.
- Check zero or negative quantities and unusual price values.
- Standardize product descriptions and categories where appropriate.

9.2 Exploratory Data Analysis

- Daily, weekly and monthly sales trends.
- Top products by units sold and revenue.
- Revenue contribution by customers and countries.
- Demand volatility by product or category.
- Price distribution and relationship between price and quantity.
- Repeat-customer patterns.
- Seasonal peaks and low-demand periods.
- Outliers and cancellation patterns.

9.3 Feature Engineering

Feature Group	Examples	Purpose
Time	Year, month, week, day of week and season proxy	Capture calendar and seasonal effects
Demand	Lag sales, rolling mean and rolling standard deviation	Forecasting
Product	Product frequency, average selling price and revenue share	Product prioritization
Customer	Recency, frequency and monetary value	Customer segmentation
Price	Current price and relative price	Demand response

10. Forecasting and Evaluation

The forecasting stage should begin with a simple baseline so that more advanced models can be compared fairly. Because sales data are time dependent, the project should use a chronological train-validation-test split rather than randomly mixing observations. Rolling-origin or walk-forward validation can be used where practical.

Metric	Why It Is Useful
MAE	Easy-to-understand average absolute forecasting error.
RMSE	Gives greater weight to large forecasting errors.
MAPE	Provides percentage error, but can be unreliable when actual demand is zero or close to zero.
WAPE	Useful for evaluating aggregate demand performance.
Bias / Mean Error	Shows whether a model consistently over-forecasts or under-forecasts.

11. Expected Findings

- Products and categories with consistently high or low demand.
- Seasonal, weekly or monthly sales patterns.
- Customer segments that contribute strongly to revenue or repeat purchases.
- Products with unstable demand and higher forecasting risk.
- Forecasting methods that perform better for different demand patterns.
- Products that require closer inventory monitoring.
- Opportunities for targeted markdowns and flexible sourcing.
- Areas where better forecasting could reduce unnecessary production and unsold inventory.

12. Strategic Recommendations

1. Introduce demand forecasting as a regular input to purchasing and production planning.
2. Use customer segmentation to support more relevant marketing and assortment decisions.
3. Create product-level risk scores based on demand, volatility and forecast confidence.
4. Plan seasonal products earlier using historical demand patterns.
5. Prioritize high-demand products to reduce avoidable stock-outs.
6. Use smaller batches or shorter planning cycles for highly unpredictable products.
7. Use price and promotion analysis to improve markdown timing.
8. Create a management dashboard showing revenue, sales, forecasts, customer segments and risk indicators.
9. In a later stage, connect the model to real inventory, supplier, lead-time and production data.
10. Use analytical insights to support sustainability by reducing overproduction and unsold stock.

13. Proposed Dashboard

- Revenue, units sold, active customers and average order value.
- Monthly demand trend with forecast.
- Top and bottom products by units and revenue.
- Demand heatmap by month and product/category.
- Customer segment size and revenue contribution.
- Demand volume versus volatility risk matrix.
- Actual versus predicted demand.
- Optional country-level sales or export analysis.

14. 34-Hour Work Plan

Activity	Planned Time
Industry research	3 hours
Problem definition	2 hours
Dataset discovery	3 hours
Data cleaning	5 hours
Exploratory analysis	4 hours
Feature engineering	3 hours
Modeling	5 hours
Evaluation	3 hours
Business insights	2 hours
Report/dashboard preparation	3 hours
Quality review	1 hour

Total planned effort: 34 hours.

15. Limitations and Ethical Considerations

The main limitation of public data is that it may not represent the exact operations of a modern Indian apparel company. The UCI Online Retail dataset is not exclusively apparel and is from 2010–2011. Some fashion datasets may also be synthetic or simulated. Public transaction datasets may not contain complete stock-on-hand, supplier lead-time, purchase-order, production-capacity, return or defect information.

- Use personally identifiable customer information only when necessary and permitted.
- Prefer aggregated customer analysis.
- Do not infer sensitive personal characteristics from purchasing behaviour.
- Clearly communicate model limitations and uncertainty.
- Keep human review in important business decisions.
- Follow the licence and attribution requirements of each dataset.

Conclusion

The research shows that the Apparel & Textiles industry has strong potential for machine learning because it generates data across sales, customers, products, production and supply chains. At the same time, the industry faces difficult problems such as changing demand, excess inventory, stock-outs, supply disruption and sustainability pressure.

Demand forecasting and inventory decision support are therefore suitable areas for this internship project. By combining historical transaction data with product, customer, price and time-related information, a data-driven framework can provide useful predictions and identify products that need attention.

The next stage of the project can involve implementing the proposed workflow using a suitable public dataset, evaluating forecasting models, developing customer and product segments, and creating a dashboard that presents the findings in a simple form for business users.

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